Cevaplar 1

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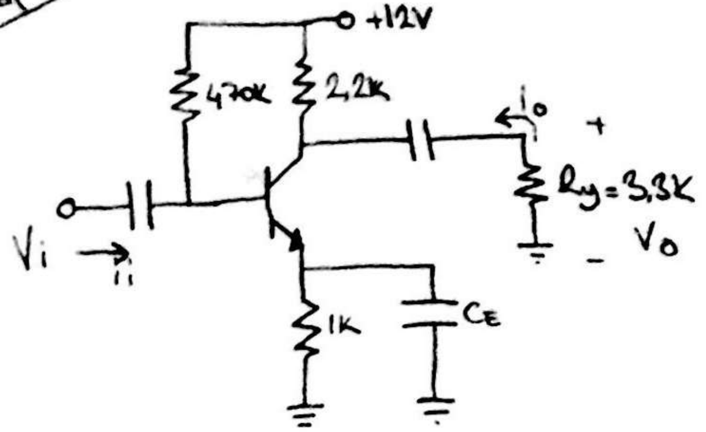
191202113



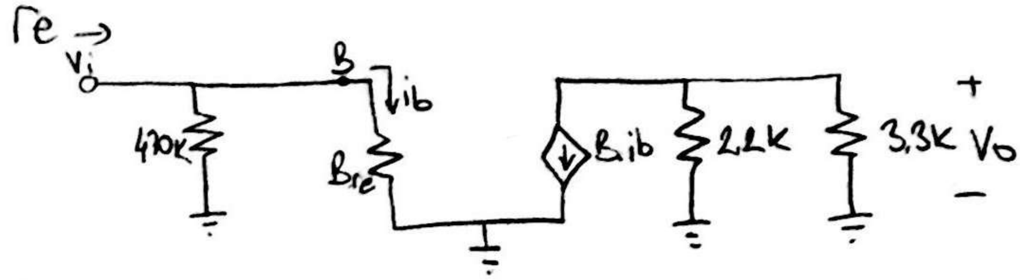
a) $r_e = ?$

b) $Z^{\circ} = ?$ (Giriş Direnci)

c) $Z_o = ?$

d) $A_V = V_o / V_i = ?$

e) $A_i = i_o / i_i = ?$



a)

$-12V + i_b \cdot 470k + 0{,}7V + I_E \cdot 1k = 0$

$-11{,}3 + i_b \cdot 470k + 151 \cdot i_b \cdot 1k = 0 \qquad I_E = (B+1) \cdot i_b = 151 \cdot i_b$

$i_b (470K + 151k) = 11{,}3$

$i_b = \frac{11{,}3}{621K} = 18{,}19\mu A \qquad I_{EDC} = i_b (B+1) = 18{,}19\mu A \cdot 151 \Rightarrow \boxed{2{,}75mA}$

$r_e = \frac{26mV}{I_{EDC}} = \frac{26mV}{2{,}75mA} = \boxed{9{,}45\Omega}$

b) $Z_i = 470K \,//\, B.r_e$

$\frac{470K \cdot 1{,}45K}{470K + 1{,}45K} \cong 1{,}445K$

c) $Z_o = \boxed{2{,}2k\Omega}$

d) $V_o = -B i_b \cdot \underbrace{(2{,}2K // 3{,}3K)}_{1{,}32K} = -150 \cdot i_b \cdot 1{,}32k$

$V_i = i_b \cdot 1{,}445K \qquad A_V = \frac{V_o}{V_i} = \frac{150 \cdot \cancel{i_b} \cdot 1{,}32K}{\cancel{i_b} \cdot 1{,}445K} = \boxed{137{,}02}$

e) $A_I = \frac{I_o}{I_1}$, $A_i = -A_V \cdot \frac{Z_i}{R_y} = -(-137{,}02) \cdot \frac{1{,}445K}{3{,}3K} \cong \boxed{59{,}99}$

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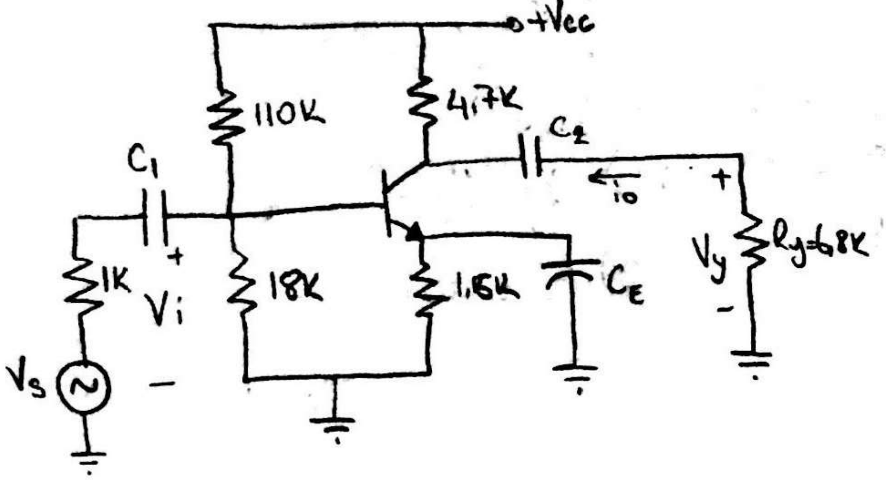


$B = h_{fe} = 200$

$h_{ie} = 1{,}2K$ ise

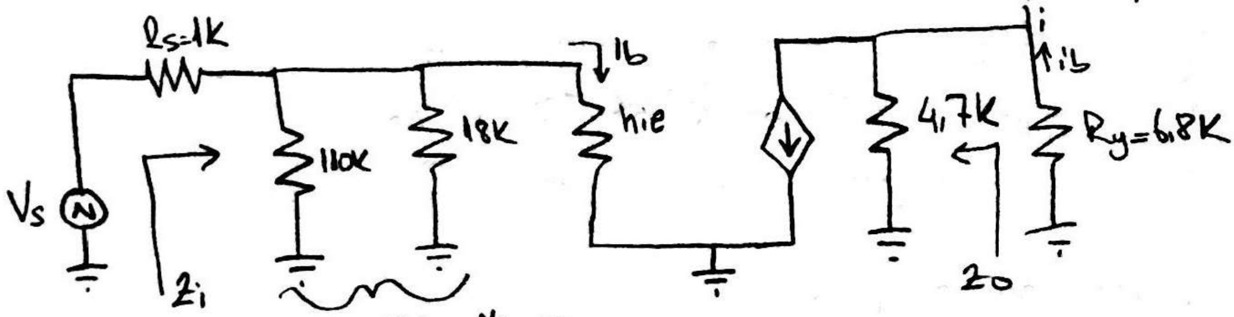
a) $Z_i = ?$

b) $Z_o = ?$

c) $A_V = \dfrac{V_y}{V_i} = ?$

d) $A_{V_S} = \dfrac{V_y}{V_s} = ?$

e) $A_i = \dfrac{i_o}{i_i} = ?$



$110K // 18K \cong 15{,}47K$

a) $Z_i = 15{,}47K // 1{,}2K = \dfrac{15{,}47 \cdot 1{,}2K}{15{,}47 + 1{,}2K} \cong \boxed{1{,}11K}$

b) $\boxed{Z_o = 4{,}7K}$

c) $V_y = -200 i_b \cdot (4{,}7K // 6{,}8K) = -200 i_b \cdot 2{,}78K$ $\qquad 4{,}7K // 6{,}8K \cong 2{,}78K$

$V_i = i_b \cdot 1{,}2K$

$A_V = \dfrac{V_y}{V_i} = \dfrac{-200 \cancel{i_b} \cdot 2{,}78K}{\cancel{i_b} \cdot 1{,}2K} \cong \boxed{-463{,}3}$

d) $A_{V_S} = \dfrac{V_y}{V_s} \Rightarrow A_{V_S} = A_V \cdot \dfrac{Z_i}{Z_i + R_s} = -463{,}3 \cdot \dfrac{1{,}11K}{1{,}11K + 1K} \cong \boxed{-243{,}7}$

e) $A_i = \dfrac{i_o}{i_i} \Rightarrow A_i = -A_V \cdot \dfrac{Z_i}{R_y} = -(-463{,}3) \cdot \dfrac{1{,}11K}{6{,}8K} \cong \boxed{75{,}63}$

Cevaplar 3)

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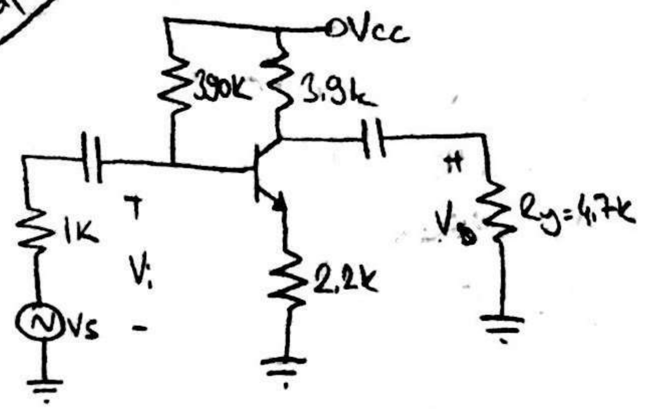


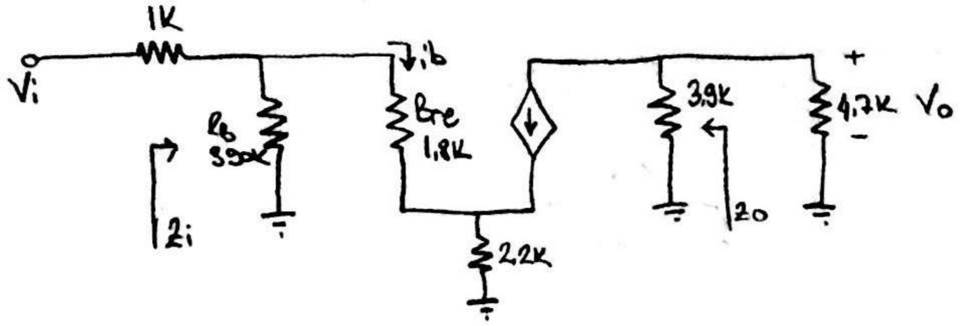
$\beta_{ac} = 150$
$r_e = 12\,\Omega$

a) Z_i, Z_o = ?

b) $A_V = \frac{V_o}{V_i} = ?$

c) $A_{V_s} = \frac{V_o}{V_s} = ?$

d) $V_s = 1mV$ sinüs ise $V_o = ?$ Çiziniz!



a) $Z_i = R_B // Z_B$

$Z_B = \beta r_e + R_E(\beta+1)$

$Z_B = 334K$

$Z_i = 390K // 334K$

$\boxed{Z_i = 179{,}91K}$ $\boxed{Z_o = 3{,}9K}$

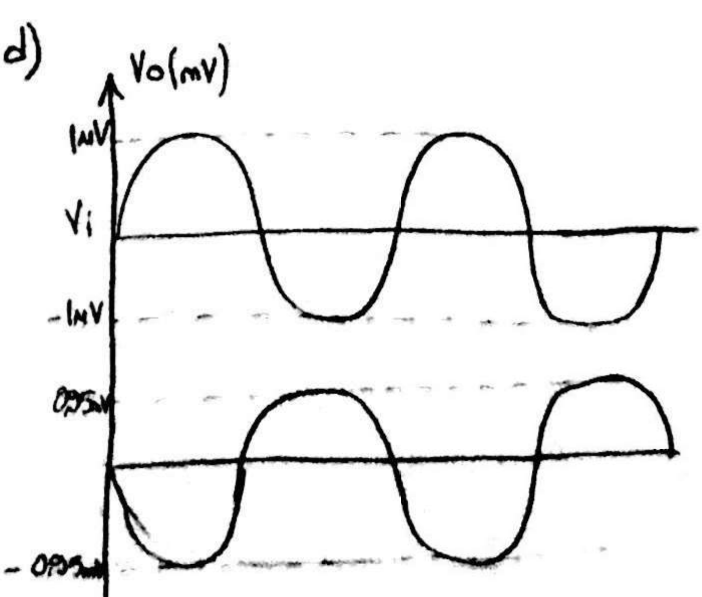
b) $V_o = -\beta i_b \underbrace{(R_C // R_y)}_{2{,}13K} = -319{,}5K \cdot i_b$

$V_i = I_1 \cdot R_B = i_b \cdot \beta r_e + i_b(\beta+1) \cdot R_E =$

$A_V = \frac{V_o}{V_i} = \frac{-319{,}5K \cdot i_b}{334K \cdot i_b} = \boxed{-0{,}956}$

c) $A_{V_s} = A_V \cdot \frac{Z_i}{R_s + Z_i} = -0{,}956 \cdot \frac{179{,}91}{180{,}91} = \boxed{-0{,}950}$

d)



$A_{V_s} = \frac{V_o}{V_s} \rightarrow -0{,}950 = \frac{V_o}{1mV}$

$\boxed{V_o = -0{,}95mV}$

Cevaplar 4

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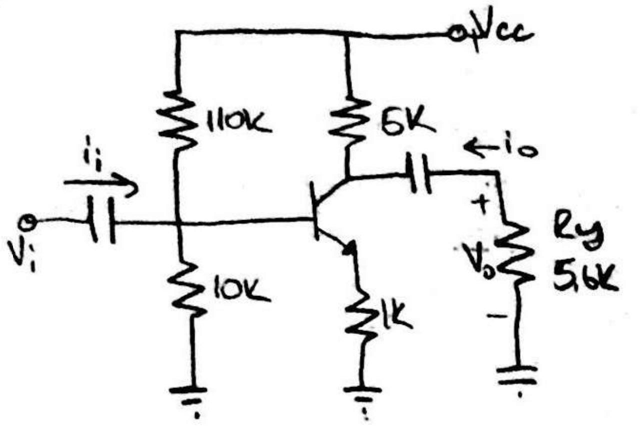


$h_{fe} = 110$

$h_{ie} = 1,1k$

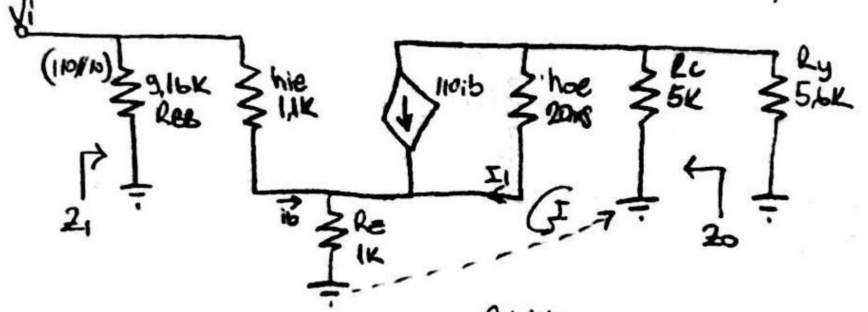
$h_{oe} = 20\mu s$

a) $Z_i = ?$

b) $Z_o = ?$

c) $A_v = \dfrac{V_o}{V_i} = ?$

d) $A_i = \dfrac{i_o}{i_i} = ?$



I göre

$$1K(i_b+I_1) + \underbrace{(5,6K // 5K)}_{2,64K} I_1 + 50k(I_1 - 110 i_b) = 0$$

$$1K i_b + 1K I_1 + 2,64 I_1 + 50K I_1 - 5500 i_b = 0$$

$$53,64K I_1 = 5499K i_b$$

$$\boxed{I_1 = 102,51 i_b}$$

a) $Z_i = R_{BB} // Z_B$ $\quad Z_B = \dfrac{V_i}{i_b}$ $\quad V_i = 1,1K i_b + 1K(i_b + I_1)$

$V_i = 104,61K i_b$

$Z_i = 110K // 10K // 104,61K$ $\quad Z_B = \dfrac{104,61K i_b}{i_b} \Rightarrow \boxed{Z_B = 104,61K}$

$\boxed{Z_i = 8,42K}$

b) $Z_o = R_c // \left[\dfrac{1}{h_{oe}} - \dfrac{h_{fe} B}{h_{oe}} + R_E(1+h_{fe})\right]$ $\quad B = -\dfrac{R_E}{R_E + [h_{ie} + R_B]} = -0,088$

$Z_o = 5K // 50k - 50 . 110 . (-0,088) + 111K] \Rightarrow 5K // 645K$

$\boxed{Z_o = 4,961K}$

c) $A_V = \dfrac{V_o}{V_i} \Rightarrow V_o = -I_1 \underbrace{(5K // 5,6)}_{2,64K} = -102,51 i_b . 2,64K = \boxed{270,6 K i_b}$

$A_V = \dfrac{270,6 K \cancel{i_b}}{104,61 K \cancel{i_b}}$ $\quad V_i = i_b . Z_B = \boxed{i_b . 104,61K}$

$\boxed{AV = 2,58}$

d) $A_i = \dfrac{I_o}{I_i}$ $\quad I_o = I_1 + \dfrac{V_o}{5K} \Rightarrow I_o = 102,61 i_b + \dfrac{(102,51 i_b . 2,64K)}{5K}$

$I_o = 48,38 i_b$

$A_i = \dfrac{48,38 \cancel{i_b}}{12,42 \cancel{i_b}}$ $\quad I_1 = i_b + \dfrac{V_i}{R_{BB}} \Rightarrow i_b + \dfrac{i_b . Z_B}{9,16K} = i_b + \dfrac{i_b . 104,61K}{9,16K}$

$\boxed{A_i = 3,89}$ $\quad I_i = 12,42 i_b$

Cevaplar 5

Ensar Azili
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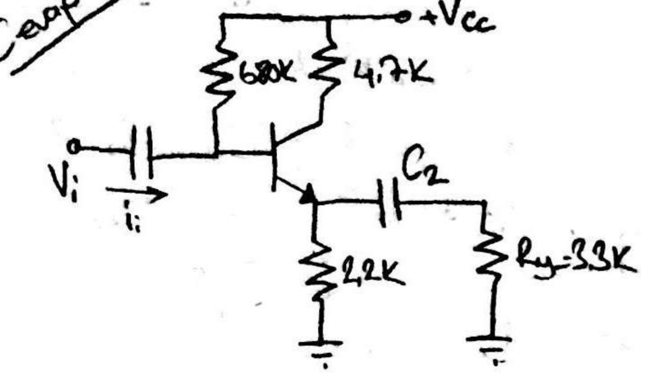


$h_{fe} = 200$

$h_{ie} = 1k$

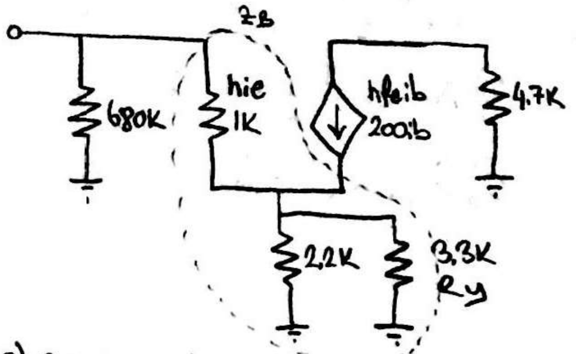
a) $Z_i = ?$

b) $Z_o = ?$

c) $A_V = \frac{V_o}{V_i} = ?$

d) $A_i = \frac{i_o}{i_i} = ?$

e) Giriş kaynak iç direnci 1K olan V_s bağlanırsa $Z_o = ?$



a) $Z_B = h_{ie} + (1+h_{fe})(R_E // R_y) = 1K + 201(1{,}32K) = 266{,}32K$

$Z_i = 680K // 266{,}32K = 191{,}37$ $\boxed{Z_i \cong 191{,}37K}$

b) $Z_o = 2{,}2K // \frac{(h_{ie}+R_B)\cdot \cancel{i_b}}{(1+h_{fe})\cdot \cancel{i_b}} = 2{,}2K // \underbrace{\frac{1+680}{201}}_{3{,}38K} = \boxed{Z_o = 1{,}33K}$

c) $A_V = \frac{V_o}{V_i} \Rightarrow V_o = 201 i_b (R_E // R_y)$

$V_o = 201 i_b (1{,}32K) = 265{,}32K\, i_b$

$V_i = i_b . Z_B = 266{,}32K\, i_b$

$A_V = \frac{265{,}32K i_b}{266{,}32K. i_b} \Rightarrow \boxed{A_V = 0{,}996}$

d) $A_i = \frac{I_o}{I_i} \Rightarrow I_o = \frac{-V_o}{3{,}3K} = \frac{265{,}32K i_b}{3{,}3K} = 80{,}4\, i_b$

$I_i = i_b + \frac{V_i}{680K} = i_b + \frac{266{,}32K. i_b}{680K} = 1{,}391\, i_b$

$A_i = \frac{80{,}4 i_b}{1{,}391 i_b} \Rightarrow \boxed{A_i = 57{,}8}$

e) $Z_o = 2{,}2K // \left(\frac{(R_s // R_B + h_{ie})}{1+h_{fe}}\right) \Rightarrow 2{,}2K // \underbrace{\frac{1K // 680K + 1K}{201}}_{0{,}00994K}$

$Z_o = 2{,}2K // 0{,}00994K$

$\boxed{Z_o = 9{,}895\Omega}$

Cevaplar 6

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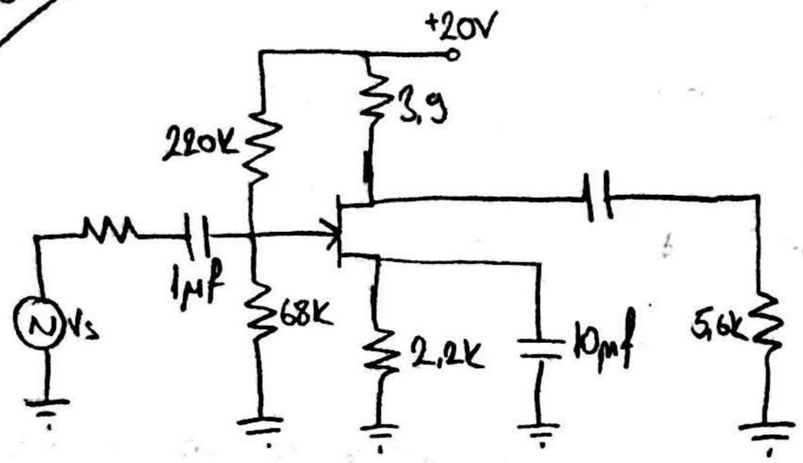


$I_{DSS} = 10mA$ $V_p = -6V$
$C_{wi} = 4pF$ $C_{wo} = 6pF$
$C_{gd} = 8pF$ $C_{gs} = 12pF$
$C_{ds} = 3pF$

Devre için;
a) DC analiz yaparak gm değeri bulunuz.
b) AC eş değerini çiziniz
c) Zi, Zo, Av, Avs değerlerini hesap.

a) $V_G = \frac{68K}{68K+220K} \cdot 20V = 4{,}72V$

$V_{GS} = V_G - I_D \cdot R_S = 4{,}72 - I_D(2{,}2K)$

$I_D = I_{DSS}\left(1 - \frac{V_{GS}}{V_p}\right)^2$

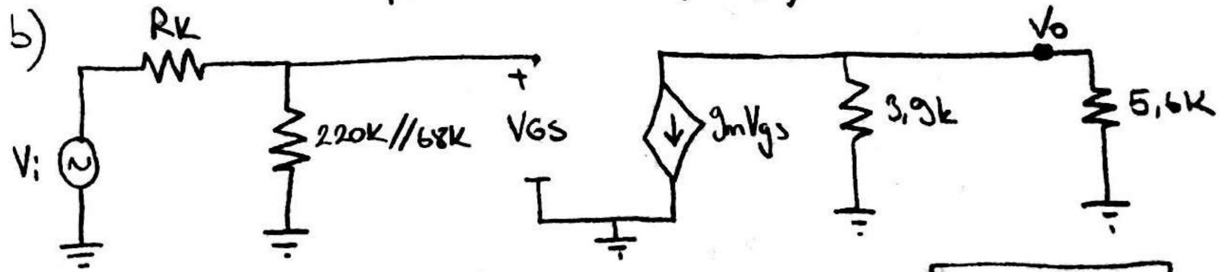
$I_D = 10\left(1 - \frac{V_{GS}}{(-6)}\right)^2 \Rightarrow V_{GSQ} = -2{,}55V$

$I_{DQ} = 3{,}3mA$

$g_{mo} = \frac{2I_{DSS}}{|V_p|} = \frac{2(10mA)}{6V} = 3{,}33mS$

$g_m = g_{mo}\left(1 - \frac{V_{GS}}{V_p}\right) = 3{,}3ms\left(1 - \frac{(-2{,}55V)}{(-6V)}\right) = \boxed{1{,}91ms}$

b)



c) $AV = -g_m(R_D // R_y) = (1{,}91ms) \cdot (3{,}9k // 5{,}6k) \Rightarrow \boxed{AV = -4{,}39}$

$AVs = AV \cdot \frac{Z_i}{Z_i + R_k} = -4{,}39 \cdot \frac{51{,}94K}{(51{,}94K) + (1{,}5K)} \Rightarrow \boxed{AVs = -4{,}27}$

$Z_i = 68K // 220K = \boxed{51{,}84K}$ $\boxed{Z_o = 3{,}9K}$

Cevaplar 7

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6. soruda verilen kuvvetlendiricinin

a) Alt kesim frekansını hesaplayınız.

b) Üst kesim frekansını hesaplayınız.

c) Kazanç – frekans eğrisini çiziniz

a) C_G için $f_G = \dfrac{1}{2\pi(R_S+Z_i)C_G} = \dfrac{1}{2\pi(1{,}5K+51{,}94K)\cdot 1\mu F} = \boxed{2{,}98\ Hz}$

C_C için $f_C = \dfrac{1}{2\pi(Z_0+R_y)C_C} = \dfrac{1}{2\pi(3{,}9K+5{,}6K)\,6{,}8\mu F} = \boxed{2{,}46\ Hz}$

C_S için $f_S = \dfrac{1}{2\pi(R_A)\cdot C_S} = \dfrac{1}{2\pi\cdot 388{,}1\cdot 10\mu F} = \boxed{41\ Hz}$

$R_A = R_K // \dfrac{1}{g_m} \Rightarrow 1{,}5K // (1/1{,}91ms) = \underline{388{,}1\,\Omega}$

$f_{alt} = \max(f_G, f_C, f_S)$

$\boxed{f_{alt} = 41 Hz}$

b) $f_H = \dfrac{1}{2\pi\cdot R_{THi}\cdot C_i}$ $\quad R_{THi} = R_K // Z_i \Rightarrow 1{,}5K // 51{,}94K = \underline{1{,}46K}$

$C_i = C_{wi} + C_{gs} + (1-A_V)C_{gd} = 4pf + 12pf + (1-(-4{,}39))\,8pf = \boxed{59{,}12}$

$f_{Hi} = \dfrac{1}{2\pi\,(1{,}46K)(59{,}12pf)} = \boxed{1{,}84\,MHz}$

$f_{Ho} = \dfrac{1}{2\pi\, R_{THo}\cdot C_o}$ $\quad R_{THo} = R_D // R_y = 3{,}9K // 5{,}6K = 2{,}3K$

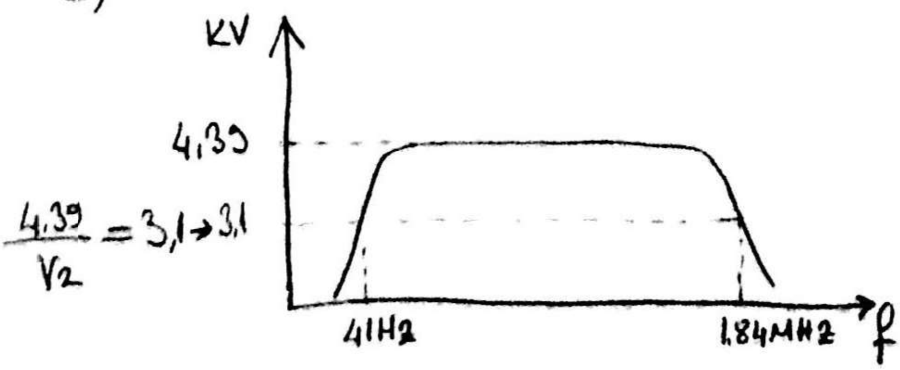
$C_o = C_{wo} + C_{ds} + C_{gd}(1 - 1/A_V) = 6pf + 3pf + (1 - 1/(-4{,}39))\cdot 8 \Rightarrow \boxed{C_o = 18{,}8}$

$f_{Ho} = \dfrac{1}{2\pi\,(2{,}3K)\,(18{,}82pf)} = \boxed{3{,}68\,MHz}$

$f_{üst} = \min(f_{Hi}, f_{Ho})$

$\boxed{f_{üst} = 1{,}84\,MHz}$

c)



Cevaplar 8

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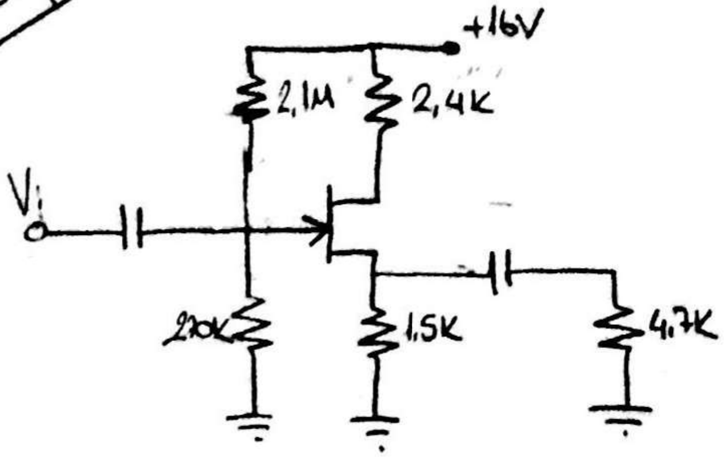
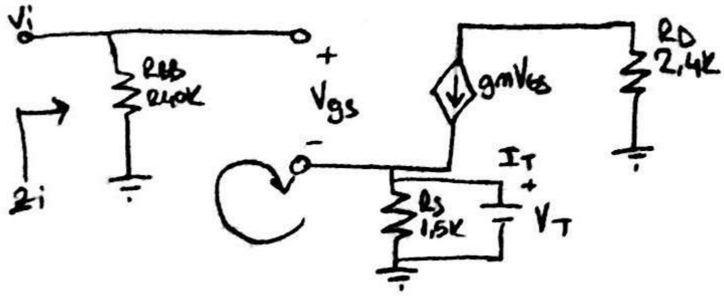


gm = 1,58 mS ise
a) Zi = ?
b) Zo = ?
c) AV değerlerini bulunuz?

a) $Z_i = 2{,}1M // 270K = \boxed{240K}$

b)



$V_T = 1{,}5K(I_T + 1{,}58m V_{gs})$

Çevre:

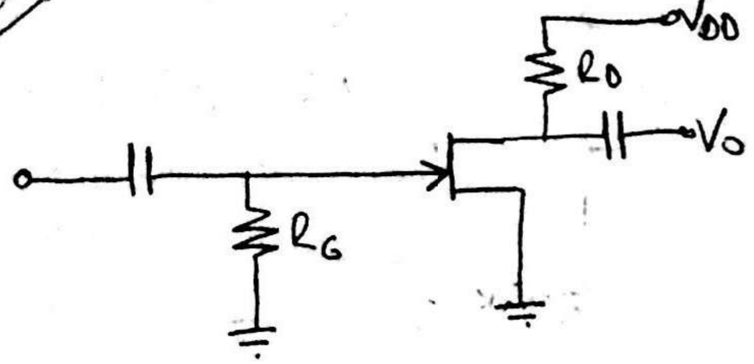
$V_{gs} + 1{,}5K I_T + 2{,}37 V_{gs} = 0$

$V_{GS} = -445 I_T \Rightarrow \boxed{Z_o = 445\,\Omega}$

c) $A_V = \dfrac{V_o}{V_i}$

$\left.\begin{array}{l} V_o = 1{,}58\ V_{gs}(1{,}5K // 4{,}7K) = 1{,}786 V_{gs} \\ V_i = V_{gs} + V_o \end{array}\right\} \boxed{A_V = 0{,}64}$

Cevaplar 9



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$V_{DD} = +22V$, $I_{DSS} = 8mA$
$V_p = -2,5V$, $r_d = 40K$
FET yükselteç devresinin kazancı -8 olacak şekilde tasarlayınız.

$$V_{GSQ} = 0V$$

$$g_m = g_{m0} = \frac{2I_{DSS}}{|V_p|} = \frac{2.(8mA)}{2,5V} = 6,4mS$$

$$A_V = -g_m (r_d \parallel R_D)$$

$$-8 = -(6,4mS)(40K \parallel R_D)$$

$$\frac{8}{6,4mS} = 1,25K = \frac{40K.R_D}{40K + R_D}$$

$$\boxed{R_D = 1,29k}$$

Cevaplar 10

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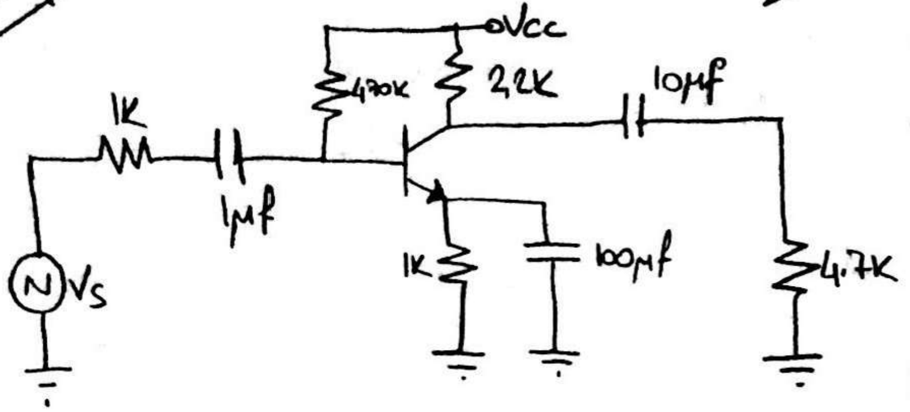

$h_{ie} = 1{,}2K$, $h_{fe} = \beta = 150$, $V_{BE} = 0{,}7V$, $C_{bc} = 4pF$, $C_{be} = 36\,pF$, $C_{ce} = 1pF$, $C_{wi} = 6pF$, $C_{wo} = 8pF$ olduğuna göre; Alt ve Üst kesim frekanslarını bulunuz.

$Z_i = 470K \,//\, 1{,}2K \qquad Z_o = 2{,}2K$

$Z_i = 1{,}196K$

$$A_V = -\frac{\beta . R_C \,//\, R_y}{h_{ie}} = \frac{-150.\overbrace{(2{,}2K \,//\, 4{,}7K)}^{1{,}49}}{1{,}2K} = \frac{-223{,}5}{1{,}2K} \Rightarrow \boxed{A_V = -186{,}25}$$

C_1 için $f_{c1} = \dfrac{1}{2\pi(1K + 1{,}196K)\,1\mu F} = 72{,}44\,Hz$

C_2 için $f_{c2} = \dfrac{1}{2\pi(2{,}2K + 4{,}7K).10\mu F} = 2{,}3\,Hz$

C_E için $R_{eş} = 1K \,//\, \underbrace{\dfrac{(1K \,//\, 470K) + 1{,}2K}{151}}_{0{,}01434} \Rightarrow R_{eş} \cong 14{,}34\,\Omega$

$f_{alt} = \max(f_{c1}, f_{c2}, f_{cE})$

$f_{cE} = \dfrac{1}{2\pi(14{,}34)\,100\mu F} = 110{,}9\,Hz$

$\boxed{f_{alt} \cong 110{,}9\,Hz}$

$C_{1\,miller} = 4pF(1 - A_V) = 753{,}28\,pF$

$C_{input} = C_{be} + C_{wi} + C_{1\,miller} = 795\,pF$

$C_{2\,miller} = 4pF\left(1 - \frac{1}{A_V}\right) \cong 4\,pF$

$C_{output} = C_{ce} + C_{wo} + C_{2\,miller} = 13\,pF$

$$f_{c\,input} = \frac{1}{2\pi(1K \,//\, 470K \,//\, 1{,}2K).795\,pF} = 367{,}44\,KHz$$

$$f_{c\,output} = \frac{1}{2\pi(2{,}2K \,//\, 4{,}7K)\,13\,pF} = 8{,}16\,MHz$$

$f_{üst} = \min(f_{c\,input}, f_{c\,output})$

$f_{üst} = 367{,}44\,KHz$